Rakesh Singh

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SUMMARY

AI Agent Engineer with 4 years of production backend experience (Java/Spring Boot, 2M daily transactions), now shipping agentic RAG: LangGraph, MCP, hybrid retrieval (pgvector + BM25 + RRF), eval-gated releases and a multi-provider LLM gateway (OpenAI, Anthropic Claude, AWS Bedrock). Author of **toolism**, an open-source token-cost optimizer for LLM tool definitions.

OPEN SOURCE

toolism — LLM tool-schema token optimizer | Python, MCP, PyPI github.com/rakeshnitb23/toolism

- Built an open-source Python library + CLI that measures and rewrites tool / MCP schemas to cut the input tokens re-sent on every agent turn; cut tool-definition tokens by **38%** across **12** public MCP servers (o200k_base + Anthropic count_tokens).
- Designed slimming as decorator-registered rule passes (redundant Pydantic titles, anyOf-null collapse, single-use $defs), each proven schema-equivalent by bidirectional property-based tests (Hypothesis).
- Shipped a GitHub Action that diffs tool-token cost against the main-branch baseline and fails PRs over budget; published to PyPI via trusted publishing; **50** stars, **16** contributors.

EXPERIENCE

AI Backend Engineer, Genpact Feb 2026 – Sep 2026

GroundedDocs — private document Q&A that answers only with faithful citations from the user's files, or abstains.

- Designed idempotent PDF ingest (PyMuPDF) with content hashing and stable chunk IDs, so changed documents re-index without duplicate vectors or stale citations; treated chunk boundaries as product decisions.
- Built tenant-scoped hybrid retrieval in PostgreSQL (pgvector + BM25 fused with RRF, bge-reranker-base cross-encoder rerank), filtered by `user_id` and workspace; **92%** recall@5 on a frozen 40-question set.
- Shipped Pydantic-validated answers with citations and calibrated abstention as the default path: **94%** citation faithfulness, **89%** / **86%** abstention precision / recall (18 answerable / 11 partial / 11 unanswerable).
- Threat-modeled with DFD + STRIDE (fail-closed auth, rate limits, secret isolation, retrieved text as untrusted); gated releases on a 30+ case adversarial suite (prompt injection/extraction, cross-user access, hostile PDFs, conflicting sources) at **97%** pass rate.
- Added a bounded LangGraph agent path only for questions hybrid RAG failed: loop limits, one read-only MCP tool, at most one ephemeral subagent; kept only where it beat deterministic RAG on quality, latency, cost and failures.
- Built eval gates for retrieval quality, citation faithfulness, abstention, security pass rate, structured-output success, latency and cost per successful task (custom harness + Ragas / DeepEval, traced in Langfuse), with version-tagged prompts and corpus.
- Built an OpenAI-compatible LLM gateway over OpenAI, Anthropic Claude, AWS Bedrock: streaming + structured outputs, tenant API keys, token budgets, retries, circuit breakers, privacy-aware caching; **$0.004** cost per successful task under failure injection.

Backend Java Developer, Genpact (Client: Shutterfly, USA) Oct 2022 – Jan 2026

- Architected Java 11 / Spring Boot microservices processing 2 million daily transactions, reducing API response times by 65%.
- Implemented Resilience4j fault tolerance across 15+ microservices, reaching 99.9% uptime; optimized PostgreSQL, MongoDB and Redis, increasing retrieval speed by 70% and cutting costs by 30%.
- Built React.js + TypeScript internal tools and a photo pipeline for 5 million daily uploads (sub-200 ms); smart compression cut storage cost by 40%, and paginated APIs over 100k+ records improved search by 60%.
- Automated release management with a TypeScript service (85% less manual coordination); mentored 3 developers, raising code coverage from 65% to 92% through TDD (JUnit 5, Mockito).

TECHNICAL SKILLS

Agentic AI	**LangGraph**, LangChain, **MCP** (servers and clients), tool calling, subagents, Pydantic structured outputs, **OpenAI**, **Anthropic Claude**, **AWS Bedrock** (Knowledge Bases, Guardrails), AgentCore
RAG	**Hybrid search** (pgvector + BM25 + RRF), cross-encoder reranking, citation grounding, agent memory (Redis, Postgres)
Evals & Observability	**Eval gates**, golden datasets, faithfulness, LLM-as-judge, **Langfuse**, Ragas, DeepEval, OpenTelemetry
LLMOps & Cost	**LLM gateway**, token budgets, caching, retries, circuit breakers, cost/successful task, model routing, toolism
AI Security	DFD + STRIDE, **prompt-injection defense**, adversarial test suites, fail-closed auth, least-privilege tools
Backend	**Python (FastAPI)**, **Java 8/11/17 (Spring Boot, Resilience4j)**, TypeScript, JavaScript, SQL, React.js
Data	**PostgreSQL**, **pgvector**, MongoDB, Redis, SQL optimization, DynamoDB
Cloud & DevOps	Docker, **Kubernetes**, **AWS** (EC2, RDS, S3, Bedrock, Lambda), Terraform, Git, CI/CD (Jenkins, GitHub Actions)
Testing & Tools	JUnit 5, Mockito, TDD, pytest, Hypothesis, Claude Code, Cursor

EDUCATION

National Institute of Technology (NIT) Bhopal Jun 2018 – Jun 2022

Bachelor of Technology (B.Tech)

ADDITIONAL

- **Leadership:** Led ISTE chapter initiatives at NIT Bhopal, organizing technical events for 200+ students; won 2 internal hackathons.
- **Community:** Volunteered 100+ hours at Arushi NGO, Bhopal, working with specially abled children.